(6) Suppose that f: $\mathbb{R}^2 \to \mathbb{R}$ is differentiable everywhere and that $y = f(x_1, x_2)$ where

$$x_1 = r\cos\theta$$

$$x_2 = r\sin\theta.$$

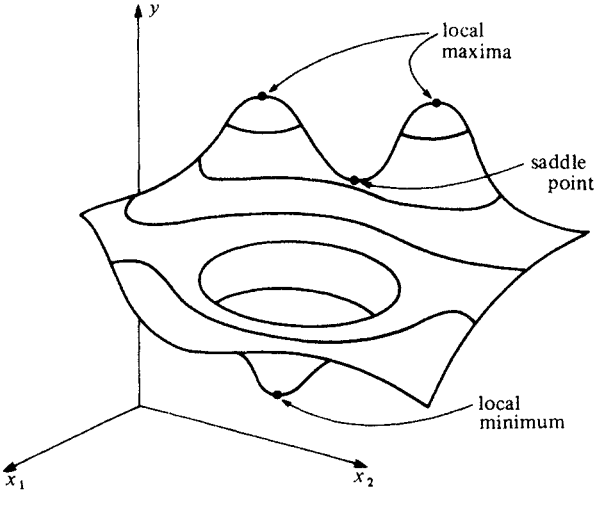
Express $\partial y/\partial x_2$ in terms of $\partial y/\partial r$ and $\partial y/\partial\theta$.

19.27 Stationary points

In §11.3 we defined a point ξ to be a *stationary point* of f if $f'(\xi) = 0$. We shall use precisely the same definition in the case of a real-valued function of a vector variable. In this case, of course, $f'(\xi)$ is a $1 \times n$ row vector and $f'(\xi) = 0$ means that all the co-ordinates of this row vector (i.e. the partial derivatives of f at ξ) are zero. In geometric terms, the fact that ξ is a stationary point means that the tangent hyperplane

$$y = f(\xi) + f'(\xi)(x - \xi)$$

reduces to $y = f(\xi)$, i.e. the tangent hyperplane is horizontal at a stationary point. The diagram shows some points on a surface $y = f(x_1, x_2)$ at which the tangent plane is horizontal.



Any local maximum or minimum at which f is differentiable is, of course, a stationary point. This follows immediately from corollary 19.9 and we quote the result as a theorem below. It is not at all uncommon, however, for stationary points in the vector case to correspond to *saddle points* as indicated in the

diagram. Less common, but not to be ignored, are other more exotic stationary points similar to those mentioned in §11.3.

19.28 Theorem Suppose that $f: A \to \mathbb{R}$ where A is a set in $\mathbb{R}^n$. If f is differentiable at $\xi \in A$ and f has a local maximum or minimum at ξ, then

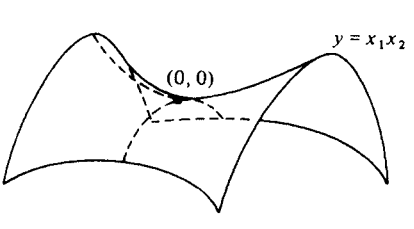
$$f'(\xi) = 0.$$

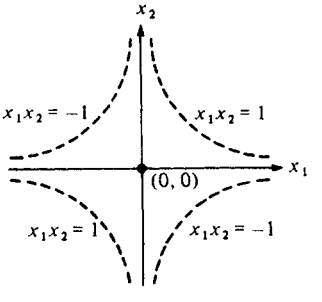
Proof This follows from corollary 19.9 and theorem 19.16.

19.29 Example Consider the function $f: \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1 x_2$. Observe that

$$f'(0, 0) = (f_1(0, 0), f_2(0, 0)) = (0, 0)$$

but f does *not* have a local maximum or minimum at (0, 0). On the line $x_1 = x_2$ we have that $f(x_1, x_2) = x_1^2 > 0$ unless $x_1 = 0$ while, on the line $x_1 = -x_2$, $f(x_1, x_2) = -x_1^2 < 0$ unless $x_1 = 0$.





The two diagrams are alternative representations of the situation. The left hand diagram makes it clear why f is said to have a saddle point at (0, 0).

19.30 Example In 'least squares analysis', one is given an *inconsistent* system $Ax = b$ of linear equations and asked to find the vector or vectors x which minimises $\|Ax - b\|$. Consider

$$y = \|Ax - b\|^2 = (Ax - b)^{\mathrm{T}}(Ax - b) = u^{\mathrm{T}}u$$

where $u = Ax - b$. By formula (V) of §19.21,

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = (2u^{\mathrm{T}}I)A = 2(Ax - b)^{\mathrm{T}}A.$$

At a local minimum, the derivative must be the zero vector and so, if x minimises $\|Ax - b\|$, then

$$2(Ax-b)^{\mathrm{T}}A = 0.$$

Recalling exercise 18.25(2), we rewrite this as

$$A^{\mathrm{T}}(Ax-b) = 0.$$

i.e.

$$A^{\mathrm{T}}Ax = A^{\mathrm{T}}b.$$

If $A^{\mathrm{T}}A$ is non-singular, it follows that there is only one candidate for a minimising x, namely

$$x = (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b.$$

In the preceding example, the geometry of the situation makes it clear that a minimising x must exist. However, a geometrical argument is no substitute for a proof and, in any case, the geometry of the situation is seldom as transparent as in example 19.30. Some method is therefore necessary for classifying stationary points into local maxima, local minima, saddle points or other more exotic categories. Our experience with the real case would indicate an examination of the *second* derivative at a stationary point. But how is this defined?

19.31 Second derivatives

We consider only the second derivative of *real-valued* functions. The reason for this is that the derivative of a real-valued function is a vector and we know how to differentiate vector-valued functions. In order to discuss second derivatives in general, we should need first to discuss the derivative of matrix-valued functions and for this purpose the notation of matrix algebra is not adequate. (See §19.48.)

Even restricting ourselves to real-valued functions, we have a small problem since §19.14 assumes that we have a column vector to differentiate whereas $f'(x)$ is a row vector. However, this problem is easily overcome. If $\phi(x)$ is a row vector we define $\phi'(x)$ by

$$\phi'(x)^{\mathrm{T}} = \frac{d}{dx}\{\phi(x)^{\mathrm{T}}\}$$

provided the latter derivative exists. If ϕ is differentiable at ξ, we then have that

$$\phi(x)-\phi(\xi)-(x-\xi)^{\mathrm{T}}\phi'(\xi) = o(x-\xi)$$

i.e. the theory for row vectors is the same as that for column vectors except that it is necessary to write $(x-\xi)^{\mathrm{T}}\phi'(\xi)$ rather than $f'(\xi)(x-\xi)$ wherever the latter expression occurs.

With this understanding, we can now define the second derivative of a real-valued function f at the point x by

$$f''(x)^{\mathrm{T}} = \frac{d}{dx} f'(x)^{\mathrm{T}}$$

provided the latter derivative exists. Since

$$f'(x)^{\mathrm{T}} = \begin{pmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{pmatrix}$$

it follows immediately from theorem 19.16 that $f''(x)$ is the $n \times n$ matrix

$$f''(x) = \begin{pmatrix} f_{11}(x) & f_{12}(x) \ldots f_{1n}(x) \\ f_{21}(x) & f_{22}(x) \ldots f_{2n}(x) \\ \ldots\ldots\ldots\ldots\ldots\ldots \\ f_{n1}(x) & f_{n2}(x) \ldots f_{nn}(x) \end{pmatrix} \tag{1}$$

where

$$f_{ij}(x) = \frac{\partial}{\partial x_i}\{f_j(x)\} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f(x)\}.$$

It is customary to abbreviate the latter notation and to write

$$f_{ij}(x) = \frac{\partial^2}{\partial x_i \partial x_j} f(x).$$

Thus, if $y = f(x)$,

$$\frac{d^2y}{dx^2} = \begin{pmatrix} \dfrac{\partial^2 y}{\partial x_1^2} & \dfrac{\partial^2 y}{\partial x_1 \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 y}{\partial x_2 \partial x_1} & \dfrac{\partial^2 y}{\partial x_2^2} \quad \cdots \dfrac{\partial^2 y}{\partial x_2 \partial x_n} \\ \vdots \\ \dfrac{\partial^2 y}{\partial x_n \partial x_1} & \dfrac{\partial^2 y}{\partial x_n \partial x_2} \cdots \dfrac{\partial^2 y}{\partial x_n^2} \end{pmatrix}$$

and so the second derivative of a real-valued function consists of an $n \times n$ matrix whose entries consist of all second order partial derivatives of the function. From theorem 19.17 we know that a sufficient condition that f be twice differentiable at a point ξ is that all these second order partial derivatives exist and are *continuous* at ξ.

19.32 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. We have that

$$\frac{\partial y}{\partial x_1} = 2x_1x_2 + x_2^3 \qquad \frac{\partial y}{\partial x_2} = x_1^2 + 3x_2^2x_1$$

and so

$$\frac{\partial^2 y}{\partial x_1^2} = 2x_2 \qquad \frac{\partial^2 y}{\partial x_1 \partial x_2} = 2x_1 + 3x_2^2$$

$$\frac{\partial^2 y}{\partial x_2 \partial x_1} = 2x_1 + 3x_2^2 \qquad \frac{\partial^2 y}{\partial x_2^2} = 6x_1x_2.$$

All these are continuous everywhere and hence f is twice differentiable at every point with

$$f''(x_1, x_2) = \begin{pmatrix} 2x_2 & 2x_1 + 3x_2^2 \\ 2x_1 + 3x_2^2 & 6x_1x_2 \end{pmatrix}.$$

It is no accident that the matrix $f''(x_1, x_2)$ in the preceding example is *symmetric*, i.e.

$$\frac{\partial^2 y}{\partial x_1 \partial x_2} = \frac{\partial^2 y}{\partial x_2 \partial x_1}.$$

19.33 Proposition If all the second order partial derivatives in the matrix (1) of §19.31 are continuous at ξ, then the $n \times n$ matrix $f''(\xi)$ is symmetric, i.e. for all i and j

$$f_{ij}(\xi) = f_{ji}(\xi).$$

19.34 Mean value theorems

The mean value theorem of §11.5 generalises immediately to the case of *real-valued* functions of a vector variable. Note, however, that it does *not* generalise to vector-valued functions. (See exercise 19.47(5).)

19.35 Theorem (mean value theorem) Let L denote the straight line segment joining $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$ and let L^0 be the set obtained by deleting a and b from L. If f is a real-valued function which is continuous at each point of L and differentiable at each point of L^0, then

$$f(b) - f(a) = f'(\xi)(b - a)$$

for some $\xi \in L^0$.

Proof Define F: $[0, 1] \to \mathbb{R}$ by

$$F(t) = f((1 - t)a + tb) \; \mathbb{R} \; (0 \leqslant t \leqslant 1).$$

Then F is continuous on $[0, 1]$ and differentiable on $(0, 1)$. (See exercise 18.20(5).) By theorem 11.6

$$\frac{F(1)-F(0)}{1-0} = F'(\tau)$$

for some $\tau \in (0, 1)$. Thus $f(b)-f(a) = F(1)-F(0) = F'(\tau)$. It remains to express $F'(\tau)$ in terms of the function f. Write $y = f(x)$ and $x = (1-t)a + tb$. Then

$$F'(t) = \frac{dy}{dt} = \frac{dy}{dx}\cdot\frac{dx}{dt} = f'(x)(b-a).$$

Writing $\xi = (1-\tau)a + \tau b$, it follows that

$$f(b)-f(a) = F'(\tau) = f'(\xi)(b-a).$$

In chapter 11, we went on to prove Taylor's theorem which, as we pointed out, may be regarded as the nth order mean value theorem. This also generalises to the case of *real-valued* functions but we consider only the case $n = 2$ since we have not discussed derivatives of higher order.

19.36 Theorem (Taylor's theorem of order 2) Suppose that f is a real-valued function which is twice differentiable at each point $x \in \mathbb{R}^n$ which satisfies $\|x-\xi\| < \Delta$. Then for any such x,

$$f(x) = f(\xi) + \frac{1}{1!}f'(\xi)(x-\xi) + \frac{1}{2!}(x-\xi)^{\mathrm{T}}f''(\eta)(x-\xi)$$

for some η on the line segment L^0 between x and ξ.

Proof We apply theorem 11.10 to the function F: $[0, 1] \to \mathbb{R}$ defined by $F(t) = f((1-t)\xi + tx)$ $(0 \leqslant t \leqslant 1)$ and obtain

$$F(1) = F(0) + \frac{1}{1!}F'(0) + \frac{1}{2!}F''(\tau)$$

for some τ satisfying $0 < \tau < 1$. We have that $F(1) = f(x)$, $F(0) = f(\xi)$ and, as in the previous theorem,

$$F'(t) = f'((1-t)\xi + tx)(x-\xi)$$

so that $F'(0) = f'(\xi)(x-\xi)$. It remains to express $F''(\tau)$ in terms of f. This requires a little care with transposition signs. (See exercise 18.25(2).) Observe that

$$F''(t)^{\mathrm{T}} = \frac{d}{dt}\{F'(t)\}^{\mathrm{T}}$$

$$= \frac{d}{dt}\{f'((1-t)\xi + tx)(x-\xi)\}^{\mathrm{T}}$$

$$= \frac{d}{dt}(x-\xi)^{\mathrm{T}} f'((1-t)\xi + tx)^{\mathrm{T}}$$

$$= (x-\xi)^{\mathrm{T}} f''((1-t)\xi + tx)^{\mathrm{T}}(x-\xi).$$

Writing $\eta = (1-\tau)\xi + tx$, we obtain that $F''(\tau)^{\mathrm{T}} = (x-\xi)^{\mathrm{T}} f''(\eta)^{\mathrm{T}}(x-\xi)$. Transposing both sides then yields $F''(\tau) = (x-\xi)^{\mathrm{T}} f''(\eta)(x-\xi)$ as required.

If ξ is a stationary point, then $f'(\xi) = 0$ and so the preceding theorem reduces to

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h$$

where η lies on the line segment between ξ and $\xi + h$. In order that ξ correspond to a local maximum it is therefore clearly sufficient that $h^{\mathrm{T}} f''(\eta) h$ be *negative* for all h of sufficiently small length. The analogous condition for a local minimum is that $h^{\mathrm{T}} f''(\eta) h$ be *positive* for all h of sufficiently small length. What properties of the matrix $f''(\eta)$ guarantee one or other of these criteria? To answer this question reasonably elegantly requires the use of rather more linear algebra than we have been assuming so far.

19.37 Eigenvalues

If M is an $n \times n$ *symmetric* matrix, there exists an orthogonal matrix H (i.e. a matrix H satisfying $H^{\mathrm{T}}H = I$) such that

$$H^{\mathrm{T}} M H = D$$

where D is a diagonal matrix, i.e.

$$D = \begin{pmatrix} \lambda_1 & 0 \ldots 0 \\ 0 & \lambda_2 \ldots 0 \\ \vdots & \\ 0 & 0 \ldots \lambda_n \end{pmatrix}$$

The entries $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal of D are called the *eigenvalues* of M and can be found by solving the polynomial equation

$$\det(M - \lambda I) = 0$$

obtained by setting the determinant of the square matrix $M - \lambda I$ equal to zero. It is a useful fact that the eigenvalues of a symmetric matrix with real entries are always real numbers.

We shall take this theory for granted. Its interest in the current context is that it allows us to discuss quadratic forms in a coherent fashion. If M is an $n \times n$ symmetric matrix and x is an $n \times 1$ column vector, we call

$$x^{\mathrm{T}} M x$$

a *quadratic form*. If $x^{\mathrm{T}}Mx > 0$ for all $x \neq 0$ we say that M is *positive definite*. If $x^{\mathrm{T}}Mx < 0$ for all $x \neq 0$ we say that M is *negative definite*. In view of the remarks at the end of §19.34, these notions are clearly relevant to the classification of stationary points.

If we write $x = Hy$, then $x^{\mathrm{T}} = y^{\mathrm{T}}H^{\mathrm{T}}$ and so

$$\begin{aligned} x^{\mathrm{T}}Mx &= y^{\mathrm{T}}H^{\mathrm{T}}MHy = y^{\mathrm{T}}Dy \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2. \end{aligned}$$

Since $\|x\|^2 = x^{\mathrm{T}}x = y^{\mathrm{T}}H^{\mathrm{T}}Hy = y^{\mathrm{T}}Iy = y^{\mathrm{T}}y = \|y\|^2$, we have that $x = 0$ if and only if $y = 0$. It follows that M is positive definite if and only if all its eigenvalues are positive. Similarly M is negative definite if and only if all its eigenvalues are negative. We may also deduce the useful inequality

$$\lambda_1\|x\|^2 \leqslant x^{\mathrm{T}}Mx \leqslant \lambda_n\|x\|^2 \tag{1}$$

where λ_1 is the smallest and λ_n is the largest of the eigenvalues of M.

If all the eigenvalues of M are non-negative, then (1) shows that $x^{\mathrm{T}}Mx$ has a minimum when $x = 0$. Similarly, if all the eigenvalues of M are non-positive, then $x^{\mathrm{T}}Mx$ has a maximum when $x = 0$. If one or more of the eigenvalues are zero, however, there are straight lines through the point 0 on which $x^{\mathrm{T}}Mx$ is identically zero. To see this we need the idea of an eigenvector.

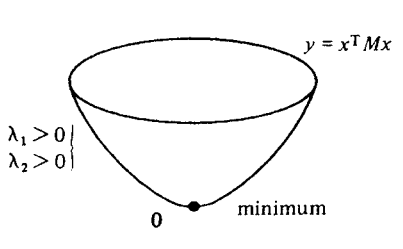
The unit vector $u^j = He^j$, where e^j is the column vector whose co-ordinates are all zero except for the jth which is equal to one, is an *eigenvector* corresponding to λ_j. If $x = tu^j$, then

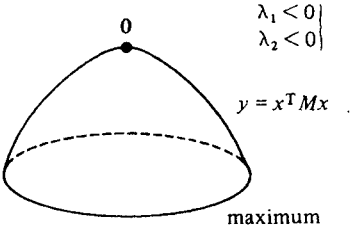
$$x^{\mathrm{T}}Mx = \lambda_j t^2. \tag{2}$$

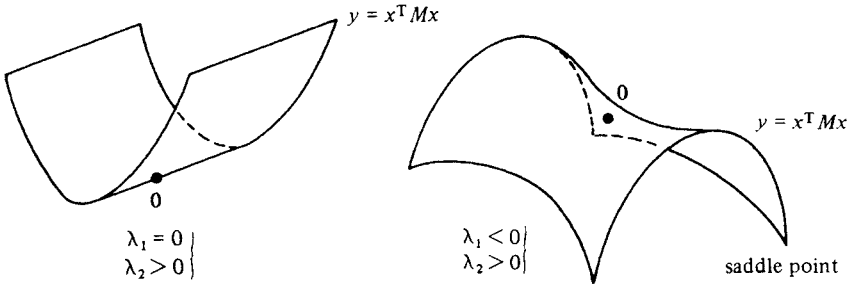
Thus, if $\lambda_j = 0$, then $x^{\mathrm{T}}Mx = 0$ for all x on the line through 0 which points in the direction of the eigenvector u^j.

It is also instructive to see what happens when some eigenvalues are positive and others are negative. Suppose in particular that $\lambda_1 < 0$ while $\lambda_n > 0$. From (2) it follows that $x^{\mathrm{T}}Mx < 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^1. Similarly, $x^{\mathrm{T}}Mx > 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^n. Thus, if we restrict our attention to values of x along one line through 0, we find that $x^{\mathrm{T}}Mx$ is strictly smallest when $x = 0$. But along a second line, $x^{\mathrm{T}}Mx$ is strictly largest when $x = 0$. Under these circumstances, we call 0 a *saddle point*.

The diagrams below illustrate the various cases we have considered.







19.38 Principal minors

If A and B are square $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$. It follows that

$$\begin{aligned}\det M &= \det(H^T DH) = (\det H^T)(\det D)(\det H) \\ &= \det(H^T H)\det D = \det I \det D.\end{aligned}$$

But the determinant of a diagonal matrix is simply the product of its diagonal entries. Hence

$$\det M = \lambda_1 \lambda_2 \ldots \lambda_n.$$

Various useful consequences follow. For example, M has a zero eigenvalue if and only if $\det M = 0$. Also $\det M$ is positive when M is positive definite and $\det M$ has the sign of $(-1)^n$ when M is negative definite. This last sentence is part of a more general result concerning the principal minors of M.

If

$$M = \begin{pmatrix} m_{11} & m_{12} & \ldots & m_{1n} \\ m_{21} & m_{22} & \ldots & m_{2n} \\ \vdots & & & \\ m_{n1} & m_{n2} & \ldots & m_{nn} \end{pmatrix}$$

then its *principal minors* are the determinants

$$P_1 = m_{11},\ P_2 = \begin{vmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{vmatrix},\ P_3 = \begin{vmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{vmatrix}$$

and so on, ending up with $P_n = \det M$. We then have the following results:

(I) The matrix M is positive definite if and only if

$$P_j > 0 \quad (j = 1, 2, \ldots, n).$$

(II) The matrix M is negative definite if and only if

$$(-1)^j P_j > 0 \quad (j = 1, 2, \ldots, n).$$

We shall take these results from linear algebra for granted. They are an extremely useful means for determining whether or not a matrix is positive or negative definite since actually to calculate the eigenvalues when $n \geqslant 3$ can be very difficult indeed. (Note incidentally that, if $P_j \geqslant 0$ $(j = 1, 2, \ldots, n)$ it does *not* follow that the eigenvalues of M are necessarily non-negative. The eigenvalues of

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

are $-1, 0$ and 2 but all the principal minors are non-negative. Nor is it true that $(-1)^j P_j \geqslant 0$ implies that the eigenvalues are necessarily non-positive.)

19.39 Examples

(i) The matrix

$$A = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}$$

is positive definite because its principal minors are 5 and

$$\begin{vmatrix} 5 & -3 \\ -3 & 5 \end{vmatrix} = 25 - 9 = 16.$$

(ii) The matrix

$$B = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

is negative definite because its principal minors are -2 and

$$\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} = 4 - 1 = 3.$$

(iii) The matrix

$$C = \begin{pmatrix} 1 & 6 \\ 6 & 4 \end{pmatrix}$$

is neither positive definite nor negative definite because

$$\det C = \begin{vmatrix} 1 & 6 \\ 6 & 4 \end{vmatrix} = 4 - 36 = -32 < 0.$$

Note that, since $\det C \neq 0$, neither eigenvalue is zero.

19.40 Classification of stationary points

Suppose that f is a real-valued function with a stationary point at ξ. If all the second order partial derivatives of f exist and are continuous at each x satisfying $\|x-\xi\|<\Delta$, it is often possible to determine whether ξ corresponds to a local maximum, a local minimum or a saddle point of f by examining the matrix $f''(\xi)$. Since f has a stationary point at ξ, $f'(\xi)=0$ and so theorem 19.36 reduces to

$$f(\xi+h)-f(\xi) = \tfrac{1}{2}h^{\mathrm{T}}f''(\eta)h$$

provided $\|h\|$ is sufficiently small. We need to relate the quadratic form $h^{\mathrm{T}}f''(\eta)h$ to the quadratic form $h^{\mathrm{T}}f''(\xi)h$. For this purpose we use the inequality

$$|h^{\mathrm{T}}Lh| \leqslant \left\{\sum_{i=1}^{n}\sum_{j=1}^{n} l_{ij}^2\right\}^{1/2}\|h\|^2$$

which holds for any $n\times n$ matrix L and any $n\times 1$ column vector h. The proof consists of two applications of the Cauchy–Schwarz inequality. We have that

$$\begin{aligned}|h^{\mathrm{T}}Lh| &= \Big|\sum_i h_i \sum_j l_{ij}h_j\Big| \\ &\leqslant \sum_i |h_i|\left\{\sum_j l_{ij}^2\right\}^{1/2}\|h\| \\ &\leqslant \left\{\sum_i\sum_j l_{ij}^2\right\}^{1/2}\|h\|^2.\end{aligned}$$

19.41 Lemma Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then, given any $\epsilon>0$, we can find a $\delta>0$ such that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h|<\epsilon\|h\|^2$$

provided that $0<\|h\|<\delta$ and η is as in theorem 19.36.

Proof We simply observe that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h| \leqslant \left\{\sum_i\sum_j (f_{ij}(\xi)-f_{ij}(\eta))^2\right\}^{1/2}\|h\|^2$$

and then use the fact that each of the second order partial derivatives f_{ij} is continuous at ξ.

19.42 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then

(i) If $f'(\xi)=0$ and $f''(\xi)$ is negative definite then f has a local maximum at ξ.

(ii) If $f'(\xi)=0$ and $f''(\xi)$ is positive definite, then f has a local minimum at ξ.

Proof We prove (ii). The proof of (i) is entirely analogous.

Let λ_1 be the smallest eigenvalue of $f''(\xi)$. It follows from (1) of §19.37 that

$$h^{\mathrm{T}} f''(\xi) h \geqslant \lambda_1 \|h\|^2$$

for all h.

Since $f''(\xi)$ is positive definite, $\lambda_1 > 0$ and so we may take $\epsilon = \frac{1}{2}\lambda_1$ in lemma 19.41. From theorem 19.36, we then obtain that

$$\begin{aligned} f(\xi + h) - f(\xi) &= \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h \\ &= \tfrac{1}{2} h^{\mathrm{T}} f''(\xi) h + \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h \\ &> \tfrac{1}{2}\lambda_1 \|h\|^2 - \tfrac{1}{2}\epsilon \|h\|^2 \end{aligned}$$

provided that $0 < \|h\| < \delta$. Hence, for $0 < \|h\| < \delta$,

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_1 \|h\|^2$$

and so f has a local minimum at ξ.

19.43 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x \in \mathbb{R}^n$ satisfying $\|x - \xi\| < \Delta$.

If $f'(\xi) = 0$ but $f''(\xi)$ is neither positive definite nor negative definite, then f has a (local) saddle point at ξ provided that

$$\det f(\xi) \neq 0.$$

Proof The conditions guarantee that the smallest eigenvalue λ_1 of $f''(\xi)$ is negative while the largest eigenvalue λ_n is positive. Using (2) of §19.37 together with the argument of theorem 19.42 we obtain that

$$f(\xi + h) - f(\xi) < \tfrac{1}{4}\lambda_1 \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^1. Similarly

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_n \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^n.

19.44 Note The condition $\det f(\xi) \neq 0$ in theorem 19.43 guarantees that no eigenvalues of $f''(\xi)$ are zero. It is worth pointing out that nothing very useful can be deduced about the behaviour of f at a stationary point ξ from the fact that an eigenvalue λ_j of $f''(\xi)$ is zero. In this case

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h$$

for sufficiently small values of h on the line through 0 in the direction u^j and we have no information about the sign of the right hand side. (When $\lambda_j \neq 0$ this is irrelevant since the term is then dominated by $\frac{1}{2}\lambda_j \|h\|^2$.)

In the case $n = 1$, $f''(\xi)$ is just a real number and theorem 19.42 is the familiar result which relates the existence of a local maximum or minimum at a stationary point with the sign of the real number $f''(\xi)$. Theorem 19.43 does not apply in the case $n = 1$ because, if $f''(\xi)$ is neither positive or negative it must be zero.

In the case $n = 2$, it is best to begin examining $f''(\xi)$ at a stationary point ξ by looking at

$$\begin{aligned}\det f''(\xi) &= \begin{vmatrix} f_{11}(\xi) & f_{12}(\xi) \\ f_{21}(\xi) & f_{22}(\xi) \end{vmatrix} \\ &= f_{11}(\xi)f_{22}(\xi) - f_{12}(\xi)^2.\end{aligned}$$

We know from §19.38 that for $f''(\xi)$ to be positive definite or negative definite it is necessary that $\det f''(\xi) > 0$. If $\det f''(\xi) > 0$ and $f_{11}(\xi) > 0$, then $f''(\xi)$ is positive definite and we have a local minimum. If $\det f''(\xi) > 0$ and $f_{11}(\xi) < 0$, then $f''(\xi)$ is negative definite and we have a local maximum. If $\det f''(\xi) < 0$, theorem 19.43 assures the existence of a saddle point.

Do *not* be tempted to draw conclusions when $\det f''(\xi) = 0$. One of the two eigenvalues is then zero and, as explained in note 19.44, the method does not work in this case. It is not possible to be tempted to draw a wrong conclusion from $\det f''(\xi) > 0$ and $f_{11}(\xi) = 0$ because the latter equation implies that $\det f''(\xi) = -f_{12}(\xi)^2$.

19.45 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^3 - 3x_1x_2^2 + x_2^4$. The stationary points are found by solving

$$\left.\begin{aligned} f_1(x_1, x_2) &= 3x_1^2 - 3x_2^2 = 0 \\ f_2(x_1, x_2) &= -6x_1x_2 + 4x_2^3 = 0 \end{aligned}\right\}$$

simultaneously. The solutions are $(3/2, 3/2)$, $(3/2, -3/2)$ and $(0, 0)$. The second derivative is

$$f''(x_1, x_2) = \begin{pmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{pmatrix}$$

and so the principal minors are

$$\det f''(x_1, x_2) = \begin{vmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{vmatrix} = 36(-x_1^2 + 2x_1x_2^2 - x_2^2)$$

and

$$f_{11}(x_1, x_2) = 6x_1.$$

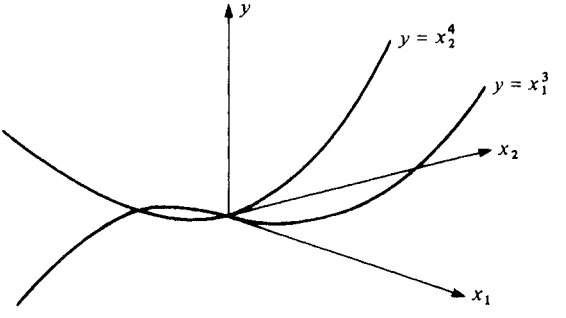
(i) Since $\det f''(3/2, 3/2) = 81 > 0$ and $f_{11}(3/2, 3/2) = 9 > 0$, we have that $(3/2, 3/2)$ is a local minimum.

(ii) Also $\det f''(3/2, -3/2) = 81 > 0$ and $f_{11}(3/2, -3/2) = 9 > 0$ and so $(3/2, -3/2)$ is also a local minimum.

(iii) Since $\det f''(0,0) = 0$, the method is useless for classifying the stationary point $(0, 0)$. Observe, however, that

$$y = f(0, x_2) = x_2^4$$
$$y = f(x_1, 0) = x_1^3$$

and so the function certainly does not have a local maximum or minimimum at $(0, 0)$.



19.46 Example Consider the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1$. The stationary points are found by solving the equations

$$\left.\begin{aligned} x_2 + x_3 &= 0 \\ x_1 + x_3 &= 0 \\ x_2 + x_1 &= 0 \end{aligned}\right\}$$

simultaneously. The only stationary point is therefore $(0, 0, 0)$. The second derivative at this point is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

which has principal minors $2, -1$ and 0. The matrix is neither positive definite nor negative definite because one of the principal minors is zero. But $f''(0, 0, 0) \neq 0$ and so we have a saddle point at $(0, 0, 0)$.

19.47 Exercise

(1) A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f(x) = (Ax + a)^{\mathrm{T}}(Bx + b)$$

where A and B are $m \times n$ matrices and x, a and b are $m \times 1$ column

vectors. Find a system of linear equations satisfied by the local maxima and minima of f. [Hint: example 19.30.]

(2) Show that that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1 e^{-x_2}$$

is twice differentiable at every point and determine $f''(x_1, x_2)$.

(3) Show that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^3 x_2}{x_1^2 + x_2^2} & (x_1, x_2) \neq (0,0) \\ 0 & (x_1, x_2) = (0,0) \end{cases}$$

has a continuous first derivative $f'\colon \mathbb{R}^2 \to \mathbb{R}^2$ at $(0,0)$ and that all its second order partial derivatives exist at $(0,0)$. Prove that

$$f_{12}(0,0) \neq f_{21}(0,0)$$

and explain why this does not contradict proposition 19.33.

(4) Classify the stationary points of the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1 - x_1 x_2$$

using the second derivative. [Hint: example 19.10.]

(5) Show that there exists no value of $\xi \in (0, 1)$ such that

$$f(1) - f(0) = f'(\xi)(1 - 0)$$

in the case of the function $f\colon \mathbb{R}^1 \to \mathbb{R}^2$ defined by

$$f(t) = \begin{pmatrix} t^2 \\ t^3 \end{pmatrix}.$$

(See §19.34.)

(6) Suppose that λ is an eigenvalue of an $n \times n$ symmetric matrix M with real entries. Using the results quoted in this chapter, prove that

$$\lambda^2 \leqslant \sum_{i=1}^{n} \sum_{j=1}^{n} m_{ij}^2.$$

19.48† Differentials

The derivative $f'(x)$ or $Df(x)$ of a vector function f at a point x was defined to be a matrix. Recall that f is differentiable at x if and only if there exists an affine function g which is tangent to f at x. The affine function g is given by

$$g(x + h) = f(x) + f'(x)h.$$